Project Title: **Hate Speech Detection through Explainable AI and Reward Optimized LLMs**

Motivation: The rapid growth of social media and online communication has increased the spread of hate speech, creating serious social and psychological impacts. Traditional hate speech detection systems often struggle with context, sarcasm, and evolving language, while their decisions are difficult to interpret. This project aims to develop an intelligent hate speech detection system using Explainable AI (XAI) to provide transparent and understandable predictions. Furthermore, Reward Optimized Large Language Models (LLMs) can improve detection accuracy by learning from feedback and aligning responses with desired outcomes. The project seeks to create a reliable, interpretable, and adaptable solution for safer and more responsible online communication.

Problem Statement: The current hate speech detection systems primarily focus on high-resource languages and text-only data. These systems are inadequate for multilingual low-resource environments where training data is limited, varied cultural expressions, frequent existence of code-mixed language, multimodal and context-dependent hate speech content. Furthermore, existing deep learning and LLM-based models often operate as black-box systems, making their predictions difficult to explain or trust. This lack of transparency is problematic in sensitive applications such as social media moderation and public safety monitoring. So, there is a strong need for an explainable, multimodal, reward-guided hate speech detection framework that supports low-resource languages, learns adaptively from feedback, provides transparent explanations, reduces bias and unfair predictions, and handles multimodal social media content effectively. Let a social media post be represented as a multimodal input: $x_i = \{T_i, V_i, C_i\}$ where T_i denotes textual content (possibly multilingual or code-mixed), V_i represents visual information (image/meme), and C_i corresponds to contextual information. The objective of hate speech detection is to learn a prediction function: $f\theta(x_i) \rightarrow y_i$ where $y_i \in \{0,1\}$ represents non-hateful or hateful content.

Proposed Scope: The project will develop a hate speech detection system using reward-optimized LLMs and Explainable AI techniques. It will classify online content, identify hateful patterns, and provide understandable explanations for predictions. The system will focus on improving accuracy, transparency, adaptability, and responsible content moderation across diverse social media text datasets.

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Deliverables Checklist:

1. Source Code (GitHub)
2. Research Paper (ACL format Paper)
3. Presentation Slides